PhenoZebra: Multimodal Latent-Space Mapping of Rare Diseases to Identify Hidden Clinical Phenotype Similarities

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Abstract

Rare disease diagnosis is challenging because clinically relevant evidence is distributed across case reports, medical images, specialty-specific descriptions, and structured phenotype vocabularies. PhenoZebra is a multimodal bioinformatics pipeline that maps rare diseases into a shared phenotype space by integrating clinical narratives, scan-set-validated clinical images, structured symptom terms, and literature-linked evidence.

Using the ZebraMap rare disease corpus, PhenoZebra indexed **69,003 valid clinical cases across 1,727 rare diseases**, including **41,954 image-linked cases across 37 medical departments**. Clinical narratives were encoded with **BioClinical-ModernBERT-large**, while image-associated information was incorporated using **MedGemma/SigLIP-derived medical visual representations**. Case-level embeddings were aggregated into **1,464 disease-level multimodal fingerprints**.

Disease fingerprints were compared using cosine similarity and contrasted against HPO-based phenotype overlap to identify candidate Clinical Cousins: disease pairs with high latent similarity but low exact ontology overlap. The pipeline processed **175,659 unique symptom strings**, generated **UMAP and network visualizations**, and exported ranked candidate pairs for downstream biological validation.

Introduction

Rare diseases often show heterogeneous and overlapping clinical presentations, making diagnosis difficult across specialties. While structured resources such as the Human Phenotype Ontology (HPO) support standardized phenotype comparison, exact ontology overlap may miss similarities expressed through narrative case descriptions, imaging findings, disease progression, or specialty-specific terminology.

PhenoZebra addresses this challenge by comparing rare diseases in two complementary ways:
Latent multimodal similarity:
Clinical narratives and image-derived features are embedded into disease-level fingerprints.

Structured phenotype similarity:
Symptom terms are mapped to HPO identifiers and compared using ontology overlap.

Central hypothesis:
Rare diseases with low exact HPO overlap may still share clinically meaningful presentation patterns in multimodal latent space.

Purpose & Research Question

Overall goal:
Develop a scalable multimodal rare-disease pipeline that integrates clinical text, image-linked case evidence, structured symptoms, and literature metadata to support phenotype-driven disease similarity discovery.

Specific objectives:
- Construct a validated rare-disease case index from ZebraMap.
- Generate multimodal disease fingerprints from clinical narratives and image-derived features.
- Compare latent disease similarity against HPO-based phenotype overlap.
- Identify and visualize candidate Clinical Cousins for downstream biological validation.

Research question:
Can multimodal embeddings reveal clinically plausible rare-disease similarities that are not captured by exact HPO term overlap?

Methods and Results

Dataset and Pipeline Overview

PhenoZebra uses **ZebraMap**, a multimodal rare-disease resource containing disease records, clinical case narratives, structured symptom terms, medical department labels, clinical image links, and literature-linked evidence. These inputs were parsed into a unified case index, where each case included disease label, department, narrative text, symptoms, image association, and source evidence.
This case index supported four downstream analysis layers: clinical text embedding, medical image-feature integration, HPO-based phenotype comparison, and disease-level similarity discovery.

ZebraMap Component	Role in PhenoZebra
Disease records	Define rare-disease entities and disease-level grouping
Case narratives	Provide clinical text for long-context embedding
Structured cases	Provide symptoms, departments, and case metadata
Clinical images	Provide image-linked phenotype evidence
Literature metadata	Link disease evidence back to source publications
Symptom terms	Support HPO mapping and ontology comparison

Metric	Value
Rare diseases loaded	**1,727**
Valid clinical cases indexed	**69,003**
Scan-set-validated image-linked cases	**41,954**
Image-linked case coverage	**60.8%**
Unique medical departments	**37**
Cases skipped due to missing text	**0**
Cases skipped due to short text	**143**

Processing

Clinical narratives were encoded with BioClinical-ModernBERT-large, while validated image-linked cases were represented using MedGemma/SigLIP-derived medical visual features. Text and image-derived embeddings were normalized, fused into multimodal case fingerprints, and aggregated across cases to generate disease-level fingerprints. In parallel, structured symptom terms were mapped to HPO identifiers to create an ontology-based phenotype baseline. Disease pairs were then compared using both cosine similarity and HPO Jaccard overlap to identify candidate Clinical Cousins, followed by UMAP visualization, network analysis, and ranked CSV/JSON export for downstream biological interpretation.

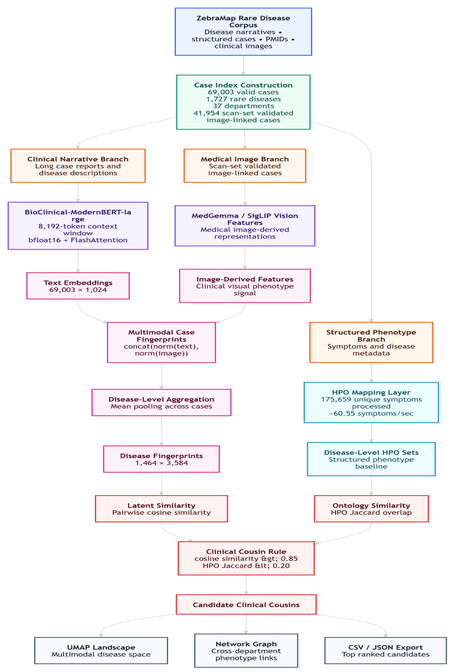


Figure 1. PhenoZebra workflow.

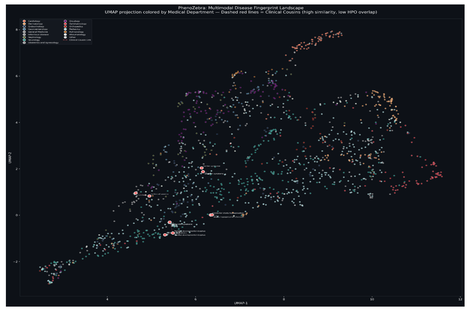


Figure 2. Multimodal disease fingerprint landscape..

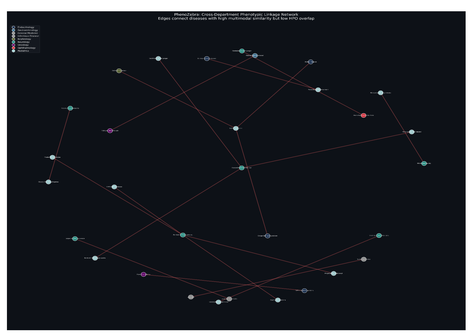


Figure 3. Cross-department Clinical Cousin network

Representative Clinical Cousin Example -
Homocystinuria due to CBS Deficiency ↔ Ornithine Transcarbamylase Deficiency

Metric	Value
Disease A	**Homocystinuria due to CBS deficiency**
Disease B	**Ornithine transcarbamylase deficiency**
Departments	**Pediatrics ↔ Neurology**
Cross-department	**Yes**
Fingerprint similarity	**0.9945**

This pair is clinically plausible because both diseases are inherited metabolic disorders that can involve the nervous system. Homocystinuria due to CBS deficiency affects methionine and homocysteine metabolism and may involve developmental delay, seizures, ocular findings, skeletal abnormalities, and thromboembolic events. Ornithine transcarbamylase deficiency affects the urea cycle and can cause hyperammonemia with encephalopathy, seizures, coma, and developmental injury.

Phenotype Layer	Homocystinuria due to CBS Deficiency	Ornithine Transcarbamylase Deficiency
Shared broad bridge	Inherited metabolic disease with CNS involvement	Inherited metabolic disease with CNS involvement
Disease-specific anchors	Ectopia lentis - **HP:0001083**; thromboembolism; skeletal abnormalities; hyperhomocysteinemia	Hyperammonemia - **HP:0001987**; urea-cycle crisis; encephalopathy; coma
Interpretation	Eye, vascular, skeletal, and homocysteine-related features dominate	Ammonia toxicity and acute neurologic crisis features dominate

Example Takeaway:
PhenoZebra identified a cross-department relationship between two metabolically distinct rare diseases that may share neurologic/metabolic presentation patterns while retaining low exact HPO overlap. This supports PhenoZebra as a hypothesis-generation framework for rare-disease phenotype candidate discovery.

Conclusions

PhenoZebra demonstrates a scalable multimodal framework for rare-disease phenotype discovery. By integrating long-context clinical text embeddings, image-derived representations, disease-level aggregation, HPO-based ontology contrast, and network visualization, the pipeline generates candidate disease relationships that may be missed by structured phenotype overlap alone.

The current execution successfully processed 69,003 clinical cases, validated 41,954 image-linked cases, generated 1,464 disease-level fingerprints, processed 175,659 unique symptom strings, and exported UMAP, network, JSON, and CSV outputs. The Homocystinuria–OTC deficiency example illustrates how PhenoZebra can surface clinically plausible cross-department candidates with high latent similarity and low exact HPO overlap. These results support PhenoZebra as a hypothesis-generation framework for rare-disease phenotype discovery.

Learnings and Future Research

Technical learnings
- Gained experience building a scalable biomedical data pipeline.
- Gained hands-on experience applying long-context biomedical language models and medical vision representations to convert unstructured clinical text.
- Learned how to evaluate AI-generated disease similarities using structured biomedical knowledge sources such as HPO, disease mechanisms, clinical phenotypes, and literature evidence

Future work
- Incorporate ontology-aware similarity - Move beyond exact HPO Jaccard overlap by using HPO hierarchy, shared ancestor terms, information content, and semantic similarity measures.
- Improve interpretability - Link each candidate pair back to representative case narratives, clinical images, symptom terms, HPO mappings, and source literature to support transparent biological interpretation.

References

- Islam MS, Jamal A, Alkhathlan A. ZebraMap: A Multimodal Rare Disease Knowledge Map with Automated Data Aggregation & LLM-Enriched Information Extraction Pipeline. Diagnostics. 2025. https://pmc.ncbi.nlm.nih.gov/articles/PMC12785374/
- Sounack T, et al. BioClinical ModernBERT: A State-of-the-Art Long-Context Encoder for Biomedical and Clinical NLP. arXiv. 2025. https://arxiv.org/abs/2506.10896
- Sellergren A, et al. MedGemma Technical Report. arXiv. 2025. https://arxiv.org/abs/2507.05201
- Gargano MA, et al. The Human Phenotype Ontology in 2024: Phenotypes Around the World. Nucleic Acids Research. 2024. https://academic.oup.com/nar/article/52/D1/D1333/7416384